

Aryan Dalal

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EDUCATION

University of California, Los Angeles (UCLA)

B.S. Computer Engineering & Mathematics

Los Angeles, CA

Expected June 2028

- Coursework: Deep Learning, Data Mining, Software Construction, Digital Design, Algorithms
- Awards: \$2000 Undergraduate Research Award, \$3200 Summer Internship Excellence Award

GPA: 3.6

WORK EXPERIENCE

NASA

Machine Learning Engineering Intern

Jun 2026 – Sep 2026

Cleveland, OH

- Built an attention-based graph neural network (GNN) in PyTorch to predict 4-port impedance spectra (501 frequency points) of randomized circuit topologies as a fast surrogate for SPICE simulation.
- Achieved common-mode R^2 of 0.986 and differential-mode R^2 of 0.978 across 100,000 simulated netlists by iterating on model architecture, normalization, and regularization using Optuna and MLflow.
- Performed readout ablations (node-level, mean, and attention pooling) and binned error diagnostics to isolate why differential-mode targets underperformed.
- Led a data-scaling study from 1K to 95K training examples using cluster-stratified sampling and a fixed held-out test set using Scikit-learn and Pandas.

RESEARCH EXPERIENCE

UCLA

Undergraduate Researcher, advised by Prof. Guido Montufar (UCLA and Max Planck Institute)

Oct 2025 – Present

Los Angeles, CA

- Studying LLM formal theorem proving (Lean, LeanDojo), autoformalization, and RLVR training through the NeurIPS 2025 MATH-AI Workshop with a graduate mentor.
- Expanded a 78-example dataset to 458 instruction pairs via Evol-Instruct-style augmentation with JSON validation.
- Fine-tuned Qwen2.5-7B-Instruct with 4-bit QLoRA for algebraic-geometry code generation.
- Built a 3-stage GEORGE pipeline (Sohoni et al.) in PyTorch and SpuCo to make a CNN robust to spurious background cues in SpuCoMNIST.
- Clustered baseline classifier outputs to find hidden subgroups without labels, then retrained on balanced batches, raising worst-group accuracy above 98%.

INVITED TALKS

- A. Dalal (July 2026). Attention-based Graph Neural Networks for Electronic Design Automation with Randomized Network Topologies. Invited talk, NASA AI Summit, Summer 2026, Cleveland, OH.

PROJECTS

Brain-Computer Interface Speech Decoding

EvalAI Brain-to-Text Benchmark 2024

Oct 2025

- Designed a unidirectional GRU decoder in PyTorch to predict phonemes from intracranial neural recordings.
- Trained with CTC loss and mixed precision on Google Cloud Platform.
- Added LayerNorm, post-GRU projection layers, and temporal masking, cutting Phoneme Error Rate from 22.7% to 19.0% at equal per-batch cost.

Skin Cancer Prediction from 3D Total Body Photographs

github.com/DalalAryan/CancerPredictionModel

Jul 2025

- Engineered 30 multi-correlation features from 10K segmented lesion images with OpenCV and scikit-learn.
- Stacked XGBoost, LightGBM, SVM, and CART base learners under an XGBoost meta-classifier.
- Reached 88.4% accuracy and 0.81 macro F1 on malignant vs. benign classification.

SKILLS

Programming: Python, C/C++, R, MATLAB, HTML/CSS, SQLite

Software Tools: NumPy, Pandas, Polars, SciPy, Git, Linux, Docker, REST APIs, FastAPI

ML/AI: PyTorch, TensorFlow, Keras, scikit-learn, Optuna, MLflow, LangChain, LangGraph, CrewAI, LlamaIndex, ChromaDB, ONNX Runtime, ptflows, vLLM, RAG